# Methods and simulations

Supplementary code documentation for:

*Multistability in species-rich communities*

Guim Aguadé-Gorgorió & Sonia Kéfi

`https://github.com/MultistabilitySynthesis`

---

This document describes the Python scripts used to produce the simulation panels in Figure 2 and the three-species HOI model and figures as discussed in the main text. Each script is self-contained and can be run independently. The codes are organized by figure panel. Figures 1 and 3 are conceptual diagrams produced without simulation code. All scripts require `numpy`, `scipy`, and `matplotlib`; `fig_HOI_C.py` additionally uses `scipy.optimize`.

---

## 1. Figure 2A — Single-species Allee effect (`fig2A.py`)

**Model**

The single-species Allee effect is described by:

$$\frac{dN}{dt} = rN\left(1 - \frac{N}{K}\right)\left(\frac{N}{A} - 1\right) \tag{1}$$

where $N$ is population abundance, $r$ the intrinsic growth rate, $K$ the carrying capacity, and $A$ the Allee threshold. This formulation produces three equilibria: $N^* = 0$ and $N^* = K$ (both stable) and $N^* = A$ (unstable), giving rise to bistability for $0 < A < K$.

**Simulation**

The ODE is integrated using `scipy.integrate.solve_ivp` with the default RK45 solver. A single trajectory is initialized above the Allee threshold ($N_0 = 0.65$) and integrated over $t \in [0, 10]$ until it approaches the high-abundance equilibrium. A pulse perturbation of $\Delta N = -0.52$ is then applied, pushing the system below the threshold, and integration continues over $t \in [10, 25]$. The dashed vertical line marks the perturbation time. Parameters: $r = 1.0$, $K = 1.0$, $A = 0.5$.

The script also contains (after the `exit()` call) code to generate a 1D phase portrait showing flow direction and equilibrium stability, which was used for reference during figure preparation.

## 2. Figure 2B — Two-species plant–pollinator model (`fig2B.py`)

**Model**

A minimal mutualistic system of one plant ($P$) and one pollinator ($A$), inspired by the model framework of Lever et al. (2014):

$$\frac{dP}{dt} = rP + P \cdot \frac{\gamma A}{1 + h\gamma A} - CP^2 + \mu \tag{2}$$

$$\frac{dA}{dt} = (r - d)A + A \cdot \frac{\gamma P}{1 + h\gamma P} - CA^2 + \mu \tag{3}$$

Mutualistic benefits saturate via a Holling type II functional response. The parameter $d$ introduces an asymmetry in intrinsic growth, representing net mortality of the pollinator in the absence of the plant. A small immigration term $\mu$ prevents exact extinction at zero and ensures the trivial equilibrium is not absorbing.

**Simulation**

Both species are initialized at $P_0 = A_0 = 0.5$ and integrated over $t \in [0, 50]$ until reaching a coexistence steady state. A pulse perturbation $\Delta A = -0.35$ is then applied to the pollinator, and the system is integrated over $t \in [50, 100]$. Depending on parameter values, the system may recover to coexistence or collapse to a low-abundance state. Parameters: $r = 0.2$, $d = 0.97$, $\gamma = 3.0$, $h = 0.5$, $C = 1.0$, $\mu = 10^{-4}$.

### 3. Figure 2C — Two-species competitive Lotka–Volterra (`fig2C.py`)

**Model**

Classical competitive Lotka–Volterra system:

$$\frac{dN_1}{dt} = r_1 N_1 \left(1 - \frac{N_1 + \alpha_{12} N_2}{K_1}\right) \tag{4}$$

$$\frac{dN_2}{dt} = r_2 N_2 \left(1 - \frac{N_2 + \alpha_{21} N_1}{K_2}\right) \tag{5}$$

When $\alpha_{12} > 1$ and $\alpha_{21} > 1$, interspecific competition exceeds intraspecific competition, producing two stable single-species equilibria and an unstable coexistence point. Initial conditions determine which species excludes the other.

**Simulation**

The system is initialized slightly asymmetrically ($N_1 = 0.4$, $N_2 = 0.6$) and integrated over $t \in [0, 75]$. A pulse perturbation of $-0.8$ is applied to the dominant species, and dynamics continue over $t \in [75, 100]$. The script also generates a phase plane with normalized flow vectors (quiver plot) to visualize the two basins of attraction. Parameters: $r_1 = r_2 = 1.0$, $K_1 = K_2 = 1.0$, $\alpha_{12} = 1.1$, $\alpha_{21} = 1.2$.

### 4. Figure 2D — Multi-species coupled Allee effect (`fig2D.py`)

**Model**

An $S$-species generalization of the single-species Allee model, where species are coupled through a facilitation matrix $\mathbf{C}$:

$$\frac{dN_i}{dt} = r_i N_i \left(1 - \frac{N_i}{K_i}\right) \left(\frac{\sum_j C_{ij} N_j}{A_i} - 1\right) \tag{6}$$

The Allee term now depends on a weighted community abundance, capturing the idea that each species benefits from the collective presence of others. When $C_{ii} = 1$ and $C_{ij} = \epsilon$ for $i \neq j$, the coupling is weak but positive, linking the bistable dynamics of all species.

**Simulation**

Five species are simulated ($S = 5$). Parameters $r_i$, $K_i$, $A_i$ are drawn from $\mathcal{N}(\mu, 0.1)$ with means matching the 1D model ($\mu_r = 1.0$, $\mu_K = 1.0$, $\mu_A = 0.5$). The cooperation matrix has diagonal entries $C_{ii} = 1$ and off-diagonal entries $C_{ij} = 0.1$. All species are initialized at $N_i = 0.65$ and integrated over $t \in [0, 10]$. A perturbation of $-4/S = -0.8$ per species is then applied and integration continues over $t \in [10, 25]$. Random seed: `numpy.random.seed(42)`.

### 5. Figure 2E — Multi-species plant–pollinator community (`fig2E.py`)

**Model**

Extension of the two-species mutualistic model (Section 2) to $n_P$ plant and $n_A$ pollinator species:

$$\frac{dP_i}{dt} = r_{P,i} P_i + P_i \cdot \frac{\sum_j \gamma_{ij} A_j}{1 + h \sum_j \gamma_{ij} A_j} - C P_i^2 + \mu \tag{7}$$

$$\frac{dA_j}{dt} = (r_{A,j} - d) A_j + A_j \cdot \frac{\sum_i \gamma_{ij} P_i}{1 + h \sum_i \gamma_{ij} P_i} - C A_j^2 + \mu \tag{8}$$

The interaction matrix $\boldsymbol{\gamma}$ is intentionally kept approximately homogeneous (not nested) for illustration; its off-diagonal entries are set to 1.2 and diagonal to 0.5. Species-specific growth rates are drawn from Uniform$(0.15, 0.25)$.

**Simulation**

Four plant and four pollinator species are simulated ($n_P = n_A = 4$). Initial abundances are drawn from Uniform$(0.4, 0.6)$. The system is integrated over $t \in [0, 30]$, after which a pulse perturbation $\Delta A_j = -0.3$ is applied to all pollinators simultaneously, and integration continues over $t \in [30, 100]$. Random seed: `numpy.random.default_rng(seed=42)`. Parameters: $d = 1.215$, $h = 0.5$, $C = 1.0$, $\mu = 10^{-4}$.

### 6. Figure 2F — Multi-species generalized Lotka–Volterra (`fig2F.py`)

**Model**

The generalized Lotka–Volterra (GLV) model with $S$ species:

$$\frac{dx_i}{dt} = x_i \left(1 - x_i + \sum_j A_{ij} x_j\right) \tag{9}$$

Interaction coefficients $A_{ij}$ are drawn from $\mathcal{N}(\mu_A, \sigma_A)$ with $A_{ii} = 0$. For sufficiently strong and heterogeneous competition ($\mu_A < 0$, $\sigma_A$ moderate), the system admits multiple alternative stable states (cliques): distinct subsets of coexisting species.

**Simulation**

A pool of $S = 50$ species is simulated. Two alternative stable communities are identified by running the system from random initial conditions until convergence, then checking whether distinct survivor sets emerge. Convergence is assessed by comparing states at $t = 1995$ and $t = 2000$; a tolerance of $10^{-4}$ is used. Once two distinct states are found, their union forms a combined system of size $N \leq S$ on which competition experiments are performed: each state is initialized as dominant while the other is present at low abundance, and a pulse perturbation (multiplicative reduction to 20%) is applied to the dominant set after $T_{\text{pre}} = 40$ time units. Post-perturbation dynamics run for $T_{\text{post}} = 20$ time units. Parameters: $\mu_A = -0.75$, $\sigma_A = 0.5$, extinction threshold $\varepsilon = 10^{-8}$. Random seed: `numpy.random.seed(42)`.

### 7. Three-species HOI model — time series and phase diagram (`fig_HOI_B.py`, `fig_HOI_C.py`)

**Overview**

Nonlinear dynamics are a well-known mechanism by which a simple system can be multistable. Yet, can internal multistability emerge in systems where species interactions are linear in their abundances? We have seen this is not the case for GLV models with interactions of the form $N_i A_{ij} N_j$, which only produce alternative assemblies. Here we study a 3-species toy model with logistic growth and a third-order interaction, in the limit-case assumption where species interact only when the full triplet is present, as in consortia in which interactions perceived by a species are mediated by two other species at the same time (Fig. 1A):

$$\frac{dN_i}{dt} = r_i N_i \left(1 - \frac{N_i - B_{ijk} N_j N_k}{K_i}\right) \tag{10}$$

where the effect on species $i$ depends on the *product* of the abundances of the other two species $j$ and $k$. This introduces a nonlinearity absent from pairwise Lotka–Volterra models.

By numerically simulating this model and finding the attractor states analytically, we find that it generates an unexpected behavior not seen in its pairwise GLV counterpart. For weak mean competition (mean$(B) > -1$), the three species coexist stably, whereas for stronger competition (mean$(B) < -2$) the triplet cannot coexist and only pairs coexist. Interestingly, there is a novel intermediate domain where both $S = 3$ and $S = 2$ states are simultaneously stable: for intermediate competition, if species start at similar abundances they can coexist. However, as soon as a competitor is pushed below a certain threshold, it can become outcompeted by the other species, leading to a shift towards an alternative stable state (Fig. 1B,C). This bistability between a three-species coexistence state and a two-species state is a property that does not arise in any pairwise sub-model.

Despite its obvious simplicity, this model provides an intuition of what it means for multistability to emerge at the scale of the model (here, three species) even when all the model components are not multistable on their own. It is a minimal example of *emergent* internal multistability, i.e. bistability that exists only at the community scale, not in any smaller subset of the model. The open question is whether this logic extends to larger communities, or whether emergent internal multistability via higher-order interactions remains confined to small motifs.

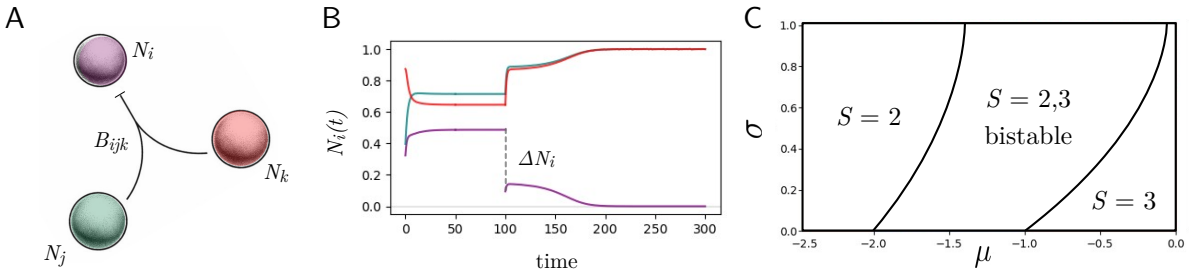



Figure 1: (A) Diagram for a three-species higher-order interaction. (B) The system reaches an $S = 3$ state, yet can shift towards $S = 2$ after a pulse perturbation, provided a species has much lower abundance than the other two (parameters: $\mu = -1.3$, $\sigma = 0.1$, $K_i = 1$ $\forall i$). (C) Phase space obtained numerically, showing an intermediate regime where bistability between $S = 3$ and $S = 2$ is possible (find in `https://github.com/GuimAguade/MultistabilitySynthesis` a detailed presentation of the numerical methods).

**Model — time series (`fig_HOI_B.py`)**

Equilibria are found analytically by solving a quintic polynomial in $N_3$, from which $N_2$ and $N_1$ are reconstructed. The system can exhibit bistability between a three-species coexistence state and a two-species state, a property that does not arise in any pairwise sub-model.

**Simulation — time series**

HOI coefficients are set to $A = -1.112$, $B = -0.908$, $C = -1.018$ (notation following the equation above), placing the system in the bistable regime identified in the phase diagram (Fig. 1C). The system is integrated in three phases: an initial transient over $t \in [0, 50]$ from a random positive initial condition; a continuation over $t \in [50, 100]$; and a targeted perturbation phase over $t \in [100, 300]$ in which species 1 is reduced to $N_1 = 0.095$, pushing it below the stability boundary and triggering a transition to the two-species state. All integrations use `scipy.integrate.solve_ivp` with RK45. Growth rates: $r_i = 1.0$ for all $i$.

**Model — phase diagram (`fig_HOI_C.py`)**

The same HOI model as above, compared against the standard pairwise GLV. For each combination of mean interaction strength $\mu$ and standard deviation $\sigma$, the script estimates via Monte Carlo sampling the probability that the system admits: (i) a feasible and stable interior equilibrium (all species present), (ii) a feasible and stable boundary equilibrium (one species absent), and (iii) both simultaneously — the condition for multistability.

**Simulation — phase diagram**

A $300 \times 300$ grid is scanned over $\mu \in [-3, 0.5]$ and $\sigma \in [0, 1.5]$. At each grid point, $n = 500$ random interaction structures are sampled. For the GLV model, the symmetric interaction matrix $A_{ij}$ is drawn from $\mathcal{N}(\mu, \sigma)$ and the interior equilibrium is solved analytically via $(\mathbf{I} - \mathbf{A})\mathbf{N} = \mathbf{1}$; boundary equilibria are solved on the $S = 2$ subsystem. Stability is assessed from the eigenvalues of the Jacobian. For the HOI model, HOI coefficients $B_i$ are drawn from $\mathcal{N}(\mu, \sigma)$; the boundary equilibrium is evaluated at $(N_1, N_2, N_3) = (0, 1, 1)$ and the interior equilibrium is found numerically using `scipy.optimize.root`. Multistability is recorded when both the interior and a boundary equilibrium are simultaneously stable. Results are stored as probability matrices and visualized as heatmaps. System size: $S = 3$.

---

**Dependencies**

All scripts were developed and tested with Python 3.10. Required packages: `numpy`, `scipy`, `matplotlib`. No installation beyond standard scientific Python is required. Scripts can be run directly from the command line (`python fig2A.py`) or from any interactive environment.

**Repository structure**

| File | Figure |
|---|---|
| `fig2A.py` | Figure 2A: single-species Allee effect |
| `fig2B.py` | Figure 2B: two-species plant–pollinator model |
| `fig2C.py` | Figure 2C: two-species competitive Lotka–Volterra |
| `fig2D.py` | Figure 2D: multi-species coupled Allee effect |
| `fig2E.py` | Figure 2E: multi-species plant–pollinator community |
| `fig2F.py` | Figure 2F: multi-species generalized Lotka–Volterra |
| `fig_HOI_B.py` | Three-species HOI model: time series |
| `fig_HOI_C.py` | Three-species HOI model: phase diagram |